Supplementary Material: When Should Robots Hide Where They Wait? Leakage-Aware Randomized Staging for Capacity-Constrained Multi-Robot Response

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S0. Reproducibility protocol

Hardware and software. One core of an Intel Xeon (2.1 GHz) cloud machine; Python 3.13, NumPy 2.5, SciPy 1.18 (HiGHS), numba 0.68, networkx 3.x; Gazebo Harmonic 8.10, ROS 2 Jazzy, Nav2 (RoboStack conda-forge builds); Mesa llvmpipe for headless rendering. All solver timings are single-threaded wall-clock, excluding Python start-up and one-time numba compilation.

Tolerances. Constraint generation adds attacks while a violation exceeds 10^{-9} (relative); defender-side certificates and the $q^\star$ tie test use 10^{-7} (relative); $\mathcal{D}^\star = \{d : \mu_d \geq P - 10^{-9}\}$; Charnes–Cooper scale cap $\lambda \leq 10^8$ (never active; largest non-tie $q^\star = 0.993$); bisection on $V^\star(q)$ to 10^{-3} in the sensitivity study.

Randomization and units. Instance generators use NumPy PCG64 with seeds fixed in the scripts (grid: `instances.py`; maps: `sim.scenario`). In the closed-loop simulation every policy evaluated for a given (layout, q) uses the same seed (CRC32 of map, layout, regime and q) and the same per-shift uniforms (configuration draw by inverse CDF, attack time, surveillance) and per-robot-slot speed and success draws, so comparisons are paired. Shifts are nested in layouts; the layout (or map, in the grid) is the independent unit, and intervals are 95% t-intervals over those units.

Lemma 2 example. Sites 1–4, $m = 2$, $c = 1$, $n = 2$, $p = 0.9$; zone A ($v = 10$) reachable from sites 1,2 and zone B ($v = 1$) from sites 3,4. Expected values under x ($\{1,2\}$ or $\{3,4\}$): $w = (2, 0)$: 5.05, $(1, 1)$: 4.95, $(0, 2)$: 10.405. Under x' ($\{1,3\}$ or $\{2,4\}$): $(2, 0)$: 1.0, $(1, 1)$: 9.9, $(0, 2)$: 10.0. Both open every site with probability $\frac{1}{2}$; $z(x) = 4.95$, $z(x') = 1.0$ (file `lemma2_example.json`).

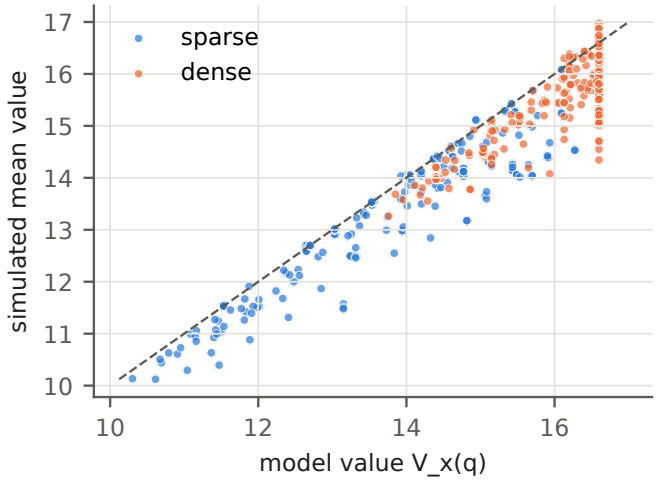


Fig. S1. Model value $V_x(q)$ versus simulated mean value for all 450 closed-loop runs ($r = 0.955$; simulator 3.1% lower on average). The dashed line marks equality.

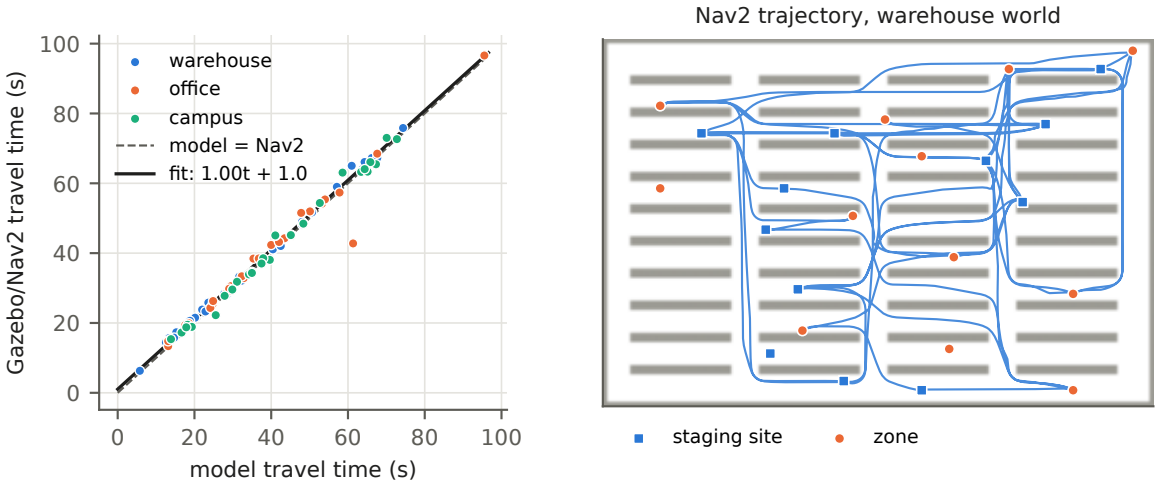


Fig. S2. Gazebo/Nav2 single-robot validation. Left: measured Nav2 travel time against the model's grid travel time for 92 site–zone legs on three worlds (fit $0.997t + 1.0\,\mathrm{s}$, $R^2 = 0.995$). Right: the recorded Nav2 trajectory of the 40-leg warehouse tour.

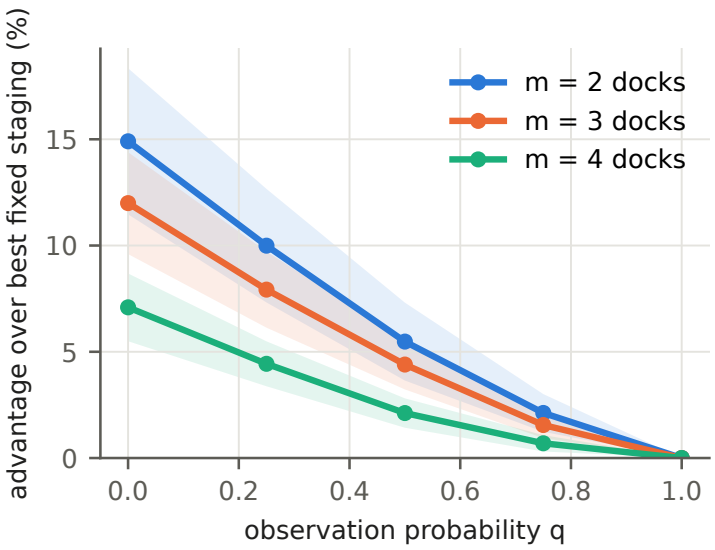


Fig. S3. Re-optimized advantage over the best fixed staging versus observation probability q (sparse, skewed, $c = 1$; mean $\pm$ s.e., 10 maps). Each instance's curve is convex and reaches zero at its own threshold $q^\star$; the plotted mean reaches zero only at $q = 1$ because $q^\star$ varies across maps.

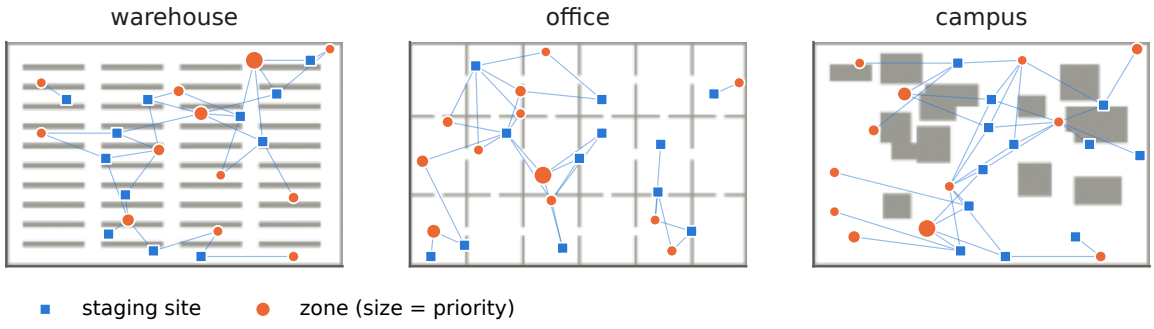


Fig. S4. Synthetic simulation maps (layout 0). Lines mark site–zone pairs reachable within the deadline along obstacle-aware shortest paths (sparse regime).

TABLE S1

CLOSED-LOOP SIMULATION, ALL POLICIES (15 LAYOUTS; MEAN SIMULATED VALUE, CHANGE VS. FIXED MAXIMIN WITH 95% t-INTERVAL OVER LAYOUTS, LAYOUTS WITH A GAIN, RELOCATION DISTANCE PER SHIFT, FRACTION OF SHIFTS OBSERVED).

regime	policy	q	value	Δ vs fixed (%)	wins	reloc. (m) / obs.
sparse	fixed maximin	0.0	13.45	0.00 ± 0.00	0/15	0 / 0.000
sparse	mexclp	0.0	13.34	-0.81 ± 1.07	0/15	0 / 0.000
sparse	uniform random	0.0	11.78	-12.02 ± 4.11	1/15	110 / 0.002
sparse	concealed q0	0.0	14.51	8.09 ± 2.67	15/15	70 / 0.001
sparse	leakage aware	0.0	14.51	8.09 ± 2.67	15/15	70 / 0.001
sparse	fixed maximin	0.25	13.46	0.00 ± 0.00	0/15	0 / 0.255
sparse	mexclp	0.25	13.36	-0.77 ± 1.08	0/15	0 / 0.255
sparse	uniform random	0.25	11.33	-15.45 ± 3.78	0/15	110 / 0.257
sparse	concealed q0	0.25	14.02	4.32 ± 2.47	12/15	70 / 0.256
sparse	leakage aware	0.25	14.21	5.76 ± 2.31	13/15	56 / 0.257
sparse	fixed maximin	0.5	13.47	0.00 ± 0.00	0/15	0 / 0.503
sparse	mexclp	0.5	13.35	-0.93 ± 1.29	0/15	0 / 0.503
sparse	uniform random	0.5	10.95	-18.43 ± 3.57	0/15	111 / 0.503
sparse	concealed q0	0.5	13.58	0.95 ± 2.41	7/15	71 / 0.503
sparse	leakage aware	0.5	13.80	2.57 ± 2.06	10/15	49 / 0.503
dense	fixed maximin	0.0	15.59	0.00 ± 0.00	0/15	0 / 0.000
dense	mexclp	0.0	15.76	1.25 ± 2.73	5/15	0 / 0.000
dense	uniform random	0.0	15.02	-3.45 ± 2.95	3/15	111 / 0.002
dense	concealed q0	0.0	15.91	2.22 ± 2.35	10/15	53 / 0.001
dense	leakage aware	0.0	15.91	2.22 ± 2.35	10/15	53 / 0.001
dense	fixed maximin	0.25	15.57	0.00 ± 0.00	0/15	0 / 0.251
dense	mexclp	0.25	15.75	1.32 ± 2.90	5/15	0 / 0.251
dense	uniform random	0.25	14.81	-4.66 ± 3.11	3/15	112 / 0.252
dense	concealed q0	0.25	15.77	1.44 ± 2.12	8/15	55 / 0.252
dense	leakage aware	0.25	15.77	1.39 ± 2.40	5/15	13 / 0.251
dense	fixed maximin	0.5	15.52	0.00 ± 0.00	0/15	0 / 0.494
dense	mexclp	0.5	15.67	1.17 ± 2.82	6/15	0 / 0.494
dense	uniform random	0.5	14.61	-5.62 ± 3.00	1/15	111 / 0.495
dense	concealed q0	0.5	15.57	0.46 ± 1.97	7/15	54 / 0.494
dense	leakage aware	0.5	15.67	1.06 ± 2.26	5/15	9 / 0.494

TABLE S2

MATRIX-FREE SOLVER STATISTICS. TOP: MAXIMUM OVER THE 9 INSTANCES OF EACH TABLE II ROW. BOTTOM: STRESS TESTS (SINGLE INSTANCES; $q \in \{0, 0.5\}$). TIME INCLUDES INITIALIZATION, ALL EXACT PURE WORST CASES μ_d NEEDED FOR P AND THE ROWS, AND ALL LISTED q; IT EXCLUDES PYTHON START-UP AND ONE-TIME NUMBA COMPILATION.

N	K	m	n	c	$\lvert\mathcal{D}\rvert$	$\lvert\mathcal{W}\rvert$	time (s)	B&B nodes / BR calls / attacks
30	20	3	6	1	1140	1.6e+06	0.36	2999 / 44 / 23
30	20	3	6	2	1140	1.6e+06	0.30	5875 / 23 / 17
40	20	3	6	1	1140	8.1e+06	0.20	4322 / 86 / 19
40	20	3	6	2	1140	8.1e+06	0.20	2911 / 36 / 14
50	20	3	6	1	1140	2.9e+07	0.62	3778 / 33 / 29
50	20	3	6	2	1140	2.9e+07	0.32	3571 / 23 / 24
30	20	3	6	1	1140	1.6e+06	9.14	445399 / 182 / 166 (uniform)
50	20	3	6	1	1140	2.9e+07	5.72	694689 / 134 / 100 (uniform)
50	20	3	8	1	1140	1.7e+09	0.18	1681 / 11 / 15 (skewed)
100	30	3	6	1	4060	1.6e+09	0.30	439 / 6 / 10 (skewed)
100	30	4	8	2	27405	3.3e+11	11.53	23005 / 62 / 25 (skewed)

TABLE S3

POLICY-DEPENDENT RELOCATION EXPOSURE (15 SPARSE LAYOUTS). EXPOSURE $= \mathbb{E}_x[\tau]/H$ OF THE EXOGENOUS OPTIMUM; GAINS VS. BEST FIXED UNDER END-TO-END EXPOSURE FOR THE EXOGENOUS OPTIMUM AND THE EXPOSURE-AWARE POLICY; LAYOUTS WHERE THE EXOGENOUS OPTIMUM FALLS BELOW FIXED; END-TO-END THRESHOLD.

H	q	exposure (%)	exogenous (%)	exposure-aware (%)	below fixed	$q^\star_{\mathrm{end}}$
8h	0.0	0.12	11.20 ± 2.80	11.20 ± 2.80	0/15	0.83
8h	0.25	0.10	7.49 ± 2.29	7.49 ± 2.29	0/15	0.83
8h	0.5	0.10	4.20 ± 1.67	4.20 ± 1.67	0/15	0.83
1h	0.0	0.93	11.06 ± 2.78	11.08 ± 2.78	0/15	0.83
1h	0.25	0.82	7.41 ± 2.27	7.41 ± 2.27	0/15	0.83
1h	0.5	0.76	4.16 ± 1.66	4.16 ± 1.66	0/15	0.83
15min	0.0	3.70	10.58 ± 2.72	10.67 ± 2.71	0/15	0.82
15min	0.25	3.30	7.14 ± 2.21	7.15 ± 2.21	0/15	0.82
15min	0.5	3.04	4.01 ± 1.62	4.02 ± 1.62	1/15	0.82
5min	0.0	11.11	9.30 ± 2.59	9.64 ± 2.57	0/15	0.81
5min	0.25	9.90	6.41 ± 2.06	6.48 ± 2.07	0/15	0.81
5min	0.5	9.13	3.61 ± 1.52	3.66 ± 1.50	2/15	0.81
2min	0.0	27.76	6.43 ± 2.37	7.56 ± 2.22	1/15	0.79
2min	0.25	24.74	4.76 ± 1.75	5.12 ± 1.78	0/15	0.79
2min	0.5	22.83	2.71 ± 1.32	2.95 ± 1.24	2/15	0.79

TABLE S4

RESPONSE-MODEL SENSITIVITY ON THE 240 GRID GAMES WITH $m \in \{3, 4\}$, $c \in \{1, 2\}$ (ALL DENSITIES AND PRIORITY PROFILES) AND ON THE FAVOURABLE CELL (SPARSE, $c = 1$, SKEWED; 20 GAMES).

	all 240				favourable cell		
variant	$G(0)$	$G(0.5)$	mean $q^\star$	gain>0	$G(0)$	$G(0.5)$	mean $q^\star$
all-or-nothing, $p = 0.7$ (paper)	2.70 ± 0.48	0.96 ± 0.20	0.48	151	9.54 ± 2.96	3.26 ± 1.39	0.71
all-or-nothing, $p = 0.5$	1.30 ± 0.24	0.50 ± 0.11	0.34	103	2.79 ± 0.87	0.63 ± 0.33	0.55
all-or-nothing, $p = 0.9$	4.93 ± 0.82	1.80 ± 0.36	0.57	173	15.40 ± 4.91	5.78 ± 2.33	0.78
partial credit, $p = 0.7$	1.69 ± 0.30	0.65 ± 0.14	0.39	122	2.06 ± 0.69	0.51 ± 0.30	0.57

TABLE S5

ONE-FACTOR-AT-A-TIME SCENARIO SENSITIVITY ON THE 15 MAP LAYOUTS (BASE: $m = 3$, $c = 1$, $n = 6$, DENSITY 0.20, ZIPF EXPONENT 1). MEAN $\pm$ 95% CI OVER LAYOUTS.

factor	level	$G(0)$ (%)	$G(0.25)$ (%)	$G(0.5)$ (%)	mean $q^\star$	gain>0
alarms n	4	3.67 ± 1.96	2.06 ± 1.32	0.96 ± 0.78	0.44	10/15
alarms n	6	11.22 ± 2.80	7.50 ± 2.29	4.21 ± 1.67	0.83	15/15
alarms n	8	14.62 ± 3.88	9.86 ± 2.98	5.46 ± 2.04	0.89	15/15
density	0.15	11.47 ± 3.02	7.68 ± 2.32	4.17 ± 1.62	0.83	15/15
density	0.2	11.22 ± 2.80	7.50 ± 2.29	4.21 ± 1.67	0.83	15/15
density	0.3	8.31 ± 3.07	5.59 ± 2.29	3.10 ± 1.52	0.85	14/15
density	0.45	3.56 ± 1.97	2.35 ± 1.37	1.23 ± 0.79	0.57	11/15
Zipf exponent	0.0	3.73 ± 2.31	2.73 ± 1.76	1.82 ± 1.17	0.42	7/15
Zipf exponent	0.5	5.27 ± 1.24	3.20 ± 0.97	1.44 ± 0.63	0.77	15/15
Zipf exponent	1.0	11.22 ± 2.80	7.50 ± 2.29	4.21 ± 1.67	0.83	15/15
Zipf exponent	2.0	28.28 ± 12.62	20.06 ± 9.71	12.60 ± 6.63	0.79	14/15
(m, c)	(3, 1)	11.22 ± 2.80	7.50 ± 2.29	4.21 ± 1.67	0.83	15/15
(m, c)	(3, 2)	3.88 ± 1.58	2.09 ± 0.91	0.61 ± 0.39	0.58	13/15
(m, c)	(4, 1)	5.10 ± 2.05	3.03 ± 1.32	1.36 ± 0.72	0.61	12/15

TABLE S6

EARLIER SAMPLED-SHIFT EXPERIMENT, SUPERSEDED BY THE EXACT-EXPECTATION RUNS OF TABLES S13–S15 (ROBOTS WITHOUT MUTUAL PERCEPTION). GAZEBO SHIFTS, WAREHOUSE LAYOUT 1, MODEL REACHABILITY, THREE NAV2 STACKS. PER SHIFT: POLICY, DISPATCHED ROBOTS, PER LEG (STATUS, ARRIVAL S / MODEL S, LATE FLAG), REALIZED VALUE, MODEL VALUE WITH THE SAME SUCCESS DRAWS, MINIMUM INTER-ROBOT DISTANCE (M).

policy	shift	movers	legs	value	model	min d
fixed	0	0	–	13.53	13.53	–
fixed	1	0	–	13.53	13.53	–
fixed	2	0	–	13.53	13.53	–
leakage aware	0	2	succ 12.0/11.6; succ 23.6/22.0	10.20	10.20	0.48
leakage aware	1	2	succ 12.0/11.6; succ 23.1/22.0	12.70	12.70	0.49
leakage aware	2	2	succ 12.0/11.6; succ 23.2/22.0	17.70	17.70	0.49
leakage aware	3	2	succ 12.0/11.6; succ 22.9/22.0	17.70	17.70	0.48
leakage aware	4	2	succ 12.0/11.6; succ 22.9/22.0	17.70	17.70	0.48
leakage aware	5	0	–	13.53	13.53	–
leakage aware	6	2	succ 12.0/11.6; succ 23.0/22.0	17.70	17.70	0.50
leakage aware	7	1	succ 32.4/31.1 late	12.70	16.03	24.04
leakage aware	8	2	succ 12.0/11.6; succ 23.0/22.0	17.70	17.70	0.48
leakage aware	9	1	succ 32.2/31.1 late	12.70	12.70	24.04
leakage aware	10	1	succ 32.2/31.1 late	12.70	16.03	24.04
leakage aware	11	1	succ 32.1/31.1 late	12.70	12.70	24.04
leakage aware	12	1	succ 32.2/31.1 late	12.70	16.03	24.04
leakage aware	13	2	succ 12.0/11.6; succ 23.1/22.0	15.20	15.20	0.48
leakage aware	14	1	succ 32.3/31.1 late	12.70	16.03	24.04
leakage aware	15	0	–	13.53	13.53	–
leakage aware	16	2	succ 12.0/11.6; succ 22.9/22.0	15.20	15.20	0.50
leakage aware	17	2	succ 12.0/11.6; succ 23.2/22.0	12.70	12.70	0.48
leakage aware	18	2	succ 12.0/11.6; succ 22.9/22.0	17.70	17.70	0.48
leakage aware	19	2	succ 12.0/11.6; succ 22.9/22.0	12.70	12.70	0.50

TABLE S7

EARLIER SAMPLED-SHIFT EXPERIMENT, SUPERSEDED BY THE EXACT-EXPECTATION RUNS OF TABLES S13–S15 (ROBOTS WITHOUT MUTUAL PERCEPTION). GAZEBO SHIFTS, OFFICE LAYOUT 0, MODEL REACHABILITY, THREE NAV2 STACKS. PER SHIFT: POLICY, DISPATCHED ROBOTS, PER LEG (STATUS, ARRIVAL S / MODEL S, LATE FLAG), REALIZED VALUE, MODEL VALUE WITH THE SAME SUCCESS DRAWS, MINIMUM INTER-ROBOT DISTANCE (M).

policy	shift	movers	legs	value	model	min d
fixed	0	0	–	14.60	14.60	–
fixed	1	0	–	14.60	14.60	–
fixed	2	0	–	14.60	14.60	–
leakage aware	0	3	succ 19.2/18.1; succ 19.7/19.3; succ 14.6/13.6	14.60	14.60	0.65
leakage aware	1	2	abor 25.8/8.7 late; succ 7.7/6.3	14.60	14.60	12.10
leakage aware	2	0	–	14.60	14.60	–
leakage aware	3	2	abor 25.0/8.7 late; succ 7.7/6.3	14.60	19.60	12.11
leakage aware	4	0	–	14.60	14.60	–
leakage aware	5	2	abor 25.4/8.7 late; succ 7.6/6.3	14.60	14.60	12.12
leakage aware	6	2	abor 25.7/8.7 late; succ 7.7/6.3	14.60	19.60	12.11
leakage aware	7	2	abor 25.3/8.7 late; succ 7.7/6.3	14.60	14.60	12.11
leakage aware	8	0	–	14.60	14.60	–
leakage aware	9	2	abor 25.7/8.7 late; succ 7.6/6.3	14.60	19.60	12.11
leakage aware	10	0	–	14.60	14.60	–
leakage aware	11	0	–	14.60	14.60	–
leakage aware	12	2	abor 25.9/8.7 late; succ 7.7/6.3	14.60	14.60	12.11
leakage aware	13	0	–	14.60	14.60	–
leakage aware	14	2	abor 25.2/8.7 late; succ 7.6/6.3	14.60	19.60	12.10
leakage aware	15	0	–	14.60	14.60	–
leakage aware	16	2	abor 25.8/8.7 late; succ 7.8/6.3	14.60	19.60	12.11
leakage aware	17	0	–	14.60	14.60	–
leakage aware	18	0	–	14.60	14.60	–
leakage aware	19	2	abor 26.1/8.7 late; succ 7.6/6.3	14.60	19.60	12.11

TABLE S8

EARLIER SAMPLED-SHIFT EXPERIMENT, SUPERSEDED BY THE EXACT-EXPECTATION RUNS OF TABLES S13–S15 (ROBOTS WITHOUT MUTUAL PERCEPTION). GAZEBO SHIFTS, WAREHOUSE LAYOUT 1, NAV2-VERIFIED REACHABILITY, NAV2 STACK OF THE DISPATCHED ROBOT ONLY. PER SHIFT: POLICY, DISPATCHED ROBOTS, PER LEG (STATUS, ARRIVAL S / MODEL S, LATE FLAG), REALIZED VALUE, MODEL VALUE WITH THE SAME SUCCESS DRAWS, MINIMUM INTER-ROBOT DISTANCE (M).

policy	shift	movers	legs	value	model	min d
fixed	0	0	–	11.03	11.03	–
fixed	1	0	–	11.03	11.03	–
fixed	2	0	–	11.03	11.03	–
leakage aware	0	0	–	10.20	10.20	–
leakage aware	1	0	–	11.03	11.03	–
leakage aware	2	1	succ 20.6/20.0	12.70	12.70	46.51
leakage aware	3	1	succ 20.6/20.0	12.70	12.70	46.51
leakage aware	4	0	–	11.03	11.03	–
leakage aware	5	0	–	11.03	11.03	–
leakage aware	6	0	–	11.03	11.03	–
leakage aware	7	1	succ 20.6/20.0	12.70	12.70	46.51
leakage aware	8	0	–	11.03	11.03	–
leakage aware	9	0	–	11.03	11.03	–
leakage aware	10	1	succ 20.6/20.0	12.70	12.70	46.53
leakage aware	11	1	succ 20.6/20.0	11.03	11.03	46.52
leakage aware	12	1	succ 20.6/20.0	12.70	12.70	46.52
leakage aware	13	0	–	10.20	10.20	–
leakage aware	14	0	–	11.03	11.03	–
leakage aware	15	0	–	11.03	11.03	–
leakage aware	16	0	–	10.20	10.20	–
leakage aware	17	1	succ 20.6/20.0	11.03	11.03	46.52
leakage aware	18	1	succ 20.6/20.0	12.70	12.70	46.53
leakage aware	19	1	succ 20.6/20.0	11.03	11.03	46.50

TABLE S9

EARLIER SAMPLED-SHIFT EXPERIMENT, SUPERSEDED BY THE EXACT-EXPECTATION RUNS OF TABLES S13–S15 (ROBOTS WITHOUT MUTUAL PERCEPTION). GAZEBO SHIFTS, OFFICE LAYOUT 0, NAV2-VERIFIED REACHABILITY, NAV2 STACK OF THE DISPATCHED ROBOT ONLY. PER SHIFT: POLICY, DISPATCHED ROBOTS, PER LEG (STATUS, ARRIVAL S / MODEL S, LATE FLAG), REALIZED VALUE, MODEL VALUE WITH THE SAME SUCCESS DRAWS, MINIMUM INTER-ROBOT DISTANCE (M).

policy	shift	movers	legs	value	model	min d
fixed	0	0	–	12.70	12.70	–
fixed	1	0	–	12.70	12.70	–
fixed	2	0	–	12.70	12.70	–
leakage aware	0	0	–	12.70	12.70	–
leakage aware	1	1	succ 7.7/6.3	17.70	17.70	34.00
leakage aware	2	0	–	12.70	12.70	–
leakage aware	3	1	succ 7.7/6.3	17.70	17.70	34.00
leakage aware	4	0	–	12.70	12.70	–
leakage aware	5	1	succ 7.7/6.3	12.70	12.70	34.00
leakage aware	6	1	succ 7.7/6.3	17.70	17.70	34.00
leakage aware	7	1	succ 7.7/6.3	17.70	17.70	34.00
leakage aware	8	0	–	12.70	12.70	–
leakage aware	9	1	succ 7.8/6.3	17.70	17.70	34.00
leakage aware	10	0	–	12.70	12.70	–
leakage aware	11	0	–	12.70	12.70	–
leakage aware	12	1	succ 7.6/6.3	17.70	17.70	34.00
leakage aware	13	0	–	12.70	12.70	–
leakage aware	14	0	–	12.70	12.70	–
leakage aware	15	0	–	12.70	12.70	–
leakage aware	16	1	succ 7.8/6.3	17.70	17.70	34.00
leakage aware	17	0	–	12.70	12.70	–
leakage aware	18	0	–	12.70	12.70	–
leakage aware	19	1	succ 7.6/6.3	17.70	17.70	34.00

TABLE S10

EARLIER SAMPLED-SHIFT EXPERIMENT, SUPERSEDED BY THE EXACT-EXPECTATION RUNS OF TABLES S13–S15 (ROBOTS WITHOUT MUTUAL PERCEPTION). GAZEBO SHIFTS, WAREHOUSE LAYOUT 1, NAV2-VERIFIED REACHABILITY, THREE NAV2 STACKS (BEHAVIOUR-TREE SERVER TIMEOUT 20 MS): ACKNOWLEDGEMENT TIMEOUTS UNDER CPU LOAD. PER SHIFT: POLICY, DISPATCHED ROBOTS, PER LEG (STATUS, ARRIVAL S / MODEL S, LATE FLAG), REALIZED VALUE, MODEL VALUE WITH THE SAME SUCCESS DRAWS, MINIMUM INTER-ROBOT DISTANCE (M).

policy	shift	movers	legs	value	model	min d
fixed	0	0	–	11.03	11.03	–
fixed	1	0	–	11.03	11.03	–
fixed	2	0	–	11.03	11.03	–
leakage aware	0	0	–	10.20	10.20	–
leakage aware	1	0	–	11.03	11.03	–
leakage aware	2	1	succ 20.6/20.0	12.70	12.70	46.53
leakage aware	3	1	succ 20.6/20.0	12.70	12.70	46.54
leakage aware	4	0	–	11.03	11.03	–
leakage aware	5	0	–	11.03	11.03	–
leakage aware	6	0	–	11.03	11.03	–
leakage aware	7	1	succ 20.6/20.0	12.70	12.70	46.52
leakage aware	8	0	–	11.03	11.03	–
leakage aware	9	0	–	11.03	11.03	–
leakage aware	10	1	abor 14.7/20.0 late	11.03	12.70	50.16
leakage aware	11	1	succ 20.5/20.0	11.03	11.03	46.54
leakage aware	12	1	abor 4.9/20.0 late	11.03	12.70	50.16
leakage aware	13	0	–	10.20	10.20	–
leakage aware	14	0	–	11.03	11.03	–
leakage aware	15	0	–	11.03	11.03	–
leakage aware	16	0	–	10.20	10.20	–
leakage aware	17	1	succ 20.6/20.0	11.03	11.03	46.54
leakage aware	18	1	abor 6.0/20.0 late	11.03	12.70	50.16
leakage aware	19	1	succ 20.5/20.0	11.03	11.03	46.54

TABLE S11

EARLIER SAMPLED-SHIFT EXPERIMENT, SUPERSEDED BY THE EXACT-EXPECTATION RUNS OF TABLES S13–S15 (ROBOTS WITHOUT MUTUAL PERCEPTION). GAZEBO SHIFTS, OFFICE LAYOUT 0, NAV2-VERIFIED REACHABILITY, THREE NAV2 STACKS (BEHAVIOUR-TREE SERVER TIMEOUT 20 MS): ACKNOWLEDGEMENT TIMEOUTS UNDER CPU LOAD. PER SHIFT: POLICY, DISPATCHED ROBOTS, PER LEG (STATUS, ARRIVAL S / MODEL S, LATE FLAG), REALIZED VALUE, MODEL VALUE WITH THE SAME SUCCESS DRAWS, MINIMUM INTER-ROBOT DISTANCE (M).

policy	shift	movers	legs	value	model	min d
fixed	0	0	–	12.70	12.70	–
fixed	1	0	–	12.70	12.70	–
fixed	2	0	–	12.70	12.70	–
leakage aware	0	0	–	12.70	12.70	–
leakage aware	1	1	succ 7.6/6.3	17.70	17.70	34.00
leakage aware	2	0	–	12.70	12.70	–
leakage aware	3	1	succ 7.8/6.3	17.70	17.70	34.00
leakage aware	4	0	–	12.70	12.70	–
leakage aware	5	1	succ 7.6/6.3	12.70	12.70	34.00
leakage aware	6	1	succ 7.6/6.3	17.70	17.70	34.00
leakage aware	7	1	abor 2.0/6.3 late	12.70	17.70	34.00
leakage aware	8	0	–	12.70	12.70	–
leakage aware	9	1	succ 7.6/6.3	17.70	17.70	34.00
leakage aware	10	0	–	12.70	12.70	–
leakage aware	11	0	–	12.70	12.70	–
leakage aware	12	1	abor 1.0/6.3 late	12.70	17.70	34.00
leakage aware	13	0	–	12.70	12.70	–
leakage aware	14	0	–	12.70	12.70	–
leakage aware	15	0	–	12.70	12.70	–
leakage aware	16	1	succ 7.7/6.3	17.70	17.70	34.00
leakage aware	17	0	–	12.70	12.70	–
leakage aware	18	0	–	12.70	12.70	–
leakage aware	19	1	succ 7.6/6.3	17.70	17.70	34.00

TABLE S12

REAL FLOOR PLAN (WILLOW GARAGE OFFICE, 5 LAYOUTS). TOP: EXACT MODEL PER LAYOUT. BOTTOM: CLOSED-LOOP CHANGE VS. BEST FIXED STAGING, MEAN $\pm$ 95% t-INTERVAL OVER THE 5 LAYOUTS (1000 PAIRED SHIFTS EACH), LAYOUTS WITH A GAIN.

layout	density	P	$G(0)$	$G(0.25)$	$G(0.5)$	$q^\star$
0	0.20	12.70	19.9	13.8	7.9	0.96
1	0.20	14.40	2.9	0.4	0.0	0.29
2	0.21	13.53	13.6	9.2	5.9	1.00
3	0.23	15.10	5.7	3.6	1.8	0.84
4	0.20	14.03	11.3	7.9	5.2	1.00

regime	policy	Δ vs fixed (%) at $q = 0$ / 0.25 / 0.5
sparse	mexclp	2.4 ± 7.1 (1/5) / 2.8 ± 8.4 (1/5) / 2.4 ± 7.1 (1/5)
sparse	uniform random	-13.6 ± 8.4 (0/5) / -16.2 ± 7.5 (0/5) / -19.6 ± 7.0 (0/5)
sparse	concealed q0	7.3 ± 9.7 (4/5) / 3.9 ± 9.4 (4/5) / -0.3 ± 8.1 (3/5)
sparse	leakage aware	7.3 ± 9.7 (4/5) / 6.6 ± 11.9 (4/5) / 5.5 ± 6.6 (4/5)
dense	mexclp	-1.0 ± 9.7 (2/5) / -1.1 ± 9.5 (1/5) / -0.4 ± 8.9 (2/5)
dense	uniform random	-4.3 ± 6.1 (1/5) / -5.5 ± 6.7 (1/5) / -5.8 ± 6.7 (1/5)
dense	concealed q0	2.7 ± 5.6 (3/5) / 1.7 ± 5.0 (2/5) / 1.1 ± 4.4 (2/5)
dense	leakage aware	2.7 ± 5.6 (3/5) / 1.0 ± 2.0 (2/5) / 0.7 ± 1.3 (2/5)

TABLE S13

CONCURRENT NAV2 EXECUTION OF EVERY DISPATCH THE POLICIES CAN PRODUCE, MODEL REACHABILITY. THREE NAV2 STACKS RUN THROUGHOUT; ROBOTS PERCEIVE EACH OTHER THROUGH SHARED POSES. VALUES ARE EXACT EXPECTATIONS OVER THE POLICY DISTRIBUTION AND ROBOT SUCCESS DRAWS, WITH NAV2 DECIDING WHICH LEGS ARE ON TIME. PER REPETITION: 79 DISPATCHES (58 WITH $\geq$2 ROBOTS MOVING, 20 WITH 3), 157 LEGS, 128 ON TIME, 6 ABORTED, 23 LATE.

layout	pairs	P (model)	fixed (Nav2)	gain model (%)	gain Nav2 (%)	failed arcs (dock,zone)
campus0	11	13.53	13.53	11.5	10.1	(0,11)
campus1	11	12.70	12.70	16.1	14.4	(11,9)
campus2	6	14.03	14.03	2.8	2.8	–
campus3	6	14.20	14.20	4.6	4.6	–
campus4	3	15.15	15.15	5.2	-2.3	(3,3)
office0	5	14.60	14.60	7.6	-0.4	(3,10), (4,10), (7,8)
office1	7	13.03	13.03	3.0	3.0	–
office2	9	12.65	12.65	11.4	11.4	–
office3	6	13.32	10.87	8.0	13.9	(7,8)
office4	8	14.77	14.77	4.9	-19.1	(4,8), (6,3)
warehouse0	7	15.43	15.43	2.2	-5.4	(7,2)
warehouse1	6	13.53	13.53	7.5	1.6	(9,5)
warehouse2	11	11.53	11.53	14.8	9.4	(2,11)
warehouse3	6	14.37	14.37	2.4	-3.4	(7,3)
warehouse4	9	12.70	12.70	10.5	8.4	(0,11)
mean $\pm$ 95% CI				7.5 ± 2.5	3.3 ± 4.9	

TABLE S14

CONCURRENT NAV2 EXECUTION OF EVERY DISPATCH THE POLICIES CAN PRODUCE, SLACK-VERIFIED REACHABILITY (SLACK ≥ 3.8 S, FIXED BEFORE THESE RUNS), EACH DISPATCH EXECUTED THREE TIMES; NAV2 GAINS ARE MEANS OVER THE THREE REPETITIONS. RUN-TO-RUN VARIABILITY: MEDIAN ARRIVAL SPREAD 0.09 S, 3 OF 87 LEGS CHANGED ON-TIME STATUS, MAXIMUM SPREAD 59 S (CONVERGING ROBOTS). THREE NAV2 STACKS RUN THROUGHOUT; ROBOTS PERCEIVE EACH OTHER THROUGH SHARED POSES. VALUES ARE EXACT EXPECTATIONS OVER THE POLICY DISTRIBUTION AND ROBOT SUCCESS DRAWS, WITH NAV2 DECIDING WHICH LEGS ARE ON TIME. PER REPETITION: 52 DISPATCHES (27 WITH $\geq$2 ROBOTS MOVING, 8 WITH 3), 87 LEGS, 80 ON TIME, 2 ABORTED, 5 LATE.

layout	pairs	P (model)	fixed (Nav2)	gain model (%)	gain Nav2 (%)	failed arcs (dock,zone)
campus0	6	12.70	12.70	7.6	7.6	–
campus1	9	12.70	12.70	16.1	10.8	(2,11), (5,11)
campus2	6	14.03	14.03	2.8	2.8	–
campus3	10	13.53	13.53	7.5	7.5	–
campus4	3	14.37	14.37	0.5	0.5	–
office0	9	12.70	12.70	12.2	-1.4	(3,10), (7,8)
office1	8	11.53	11.53	3.3	3.3	–
office2	9	12.65	12.65	9.1	9.1	–
office3	9	11.87	11.87	2.0	2.0	–
office4	6	13.53	13.53	4.2	4.2	–
warehouse0	5	14.53	14.53	6.3	6.3	–
warehouse1	7	11.03	11.03	2.6	2.6	–
warehouse2	4	11.53	11.53	1.5	1.5	–
warehouse3	1	13.53	13.53	0.0	0.0	–
warehouse4	6	12.65	10.20	3.2	27.9	(0,10)
mean $\pm$ 95% CI				5.3 ± 2.5	5.7 ± 3.9	

TABLE S15

CONCURRENT NAV2 EXECUTION OF EVERY DISPATCH THE POLICIES CAN PRODUCE, CONCURRENCY-AWARE REACHABILITY (ADDITIONALLY, FOR ZONES SERVED BY ≥ 2 ROBOTS, SLACK ≥ 9.66 S: THE 95TH PERCENTILE OF SHARED-ZONE ARRIVAL DELAYS IN THE MODEL-REACHABILITY ROUND, COMPUTED ON THE 14 LAYOUTS COMPLETE WHEN THIS ROUND WAS PLANNED; 9.6 S ON ALL 15). THREE NAV2 STACKS RUN THROUGHOUT; ROBOTS PERCEIVE EACH OTHER THROUGH SHARED POSES. VALUES ARE EXACT EXPECTATIONS OVER THE POLICY DISTRIBUTION AND ROBOT SUCCESS DRAWS, WITH NAV2 DECIDING WHICH LEGS ARE ON TIME. PER REPETITION: 22 DISPATCHES (10 WITH $\geq$2 ROBOTS MOVING, 3 WITH 3), 35 LEGS, 32 ON TIME, 1 ABORTED, 2 LATE.

layout	pairs	P (model)	fixed (Nav2)	gain model (%)	gain Nav2 (%)	failed arcs (dock,zone)
campus0	1	12.70	12.70	0.0	0.0	–
campus1	9	12.70	12.70	16.1	7.0	(2,11)
campus2	6	12.78	12.78	4.2	4.2	–
campus3	7	13.53	13.53	4.8	4.8	–
campus4	2	12.53	12.53	0.0	0.0	–
office0	3	12.70	12.70	9.6	0.0	(7,8)
office1	3	11.10	11.10	3.0	3.0	–
office2	8	12.65	12.65	6.9	6.9	–
office3	2	10.20	10.20	0.0	0.0	–
office4	3	12.37	12.37	1.8	1.8	–
warehouse0	1	12.70	12.70	0.0	0.0	–
warehouse1	8	11.03	11.03	2.6	2.6	–
warehouse2	1	10.20	10.20	0.0	0.0	–
warehouse3	1	12.53	12.53	0.0	0.0	–
warehouse4	3	12.65	12.65	0.2	0.2	–
mean $\pm$ 95% CI				3.3 ± 2.5	2.0 ± 1.4	